

Time, E8, and the Universal Learning Structure

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E8 Lie Group: The Largest Exceptional Symmetry

[Conway-Sloane 1999; Viazovska 2017; Adams 1996; Goddard et al 1972]

DIMENSION: 248. RANK: 8. ROOTS: 240.

WEYL GROUP ORDER: 696729600. COXETER NUMBER: 30.

EXPONENTS: [1, 7, 11, 13, 17, 19, 23, 29]

DEGREES: [2, 8, 12, 14, 18, 20, 24, 30]

KEY FACT: The non-trivial exponents (7,11,13,17,19,23,29) are exactly the first 7 PRIMES after 5.

ROOT PARTITION:

112 roots = D8 (bosonic sector)

128 roots = half-spin (fermionic sector)

240 = 112 + 128 (E8 = bosonic + fermionic)

This is the 0/0 of SUPERSTRING THEORY:

Bosonic string: 26 dimensions

Superstring: 10 dimensions

Compactified: $26 - 10 = 16 = 2 \times 8$ (the E8 lattice)

DENSEST PACKING [Viazovska 2017]:

The E8 lattice is the densest sphere packing in $\mathbb{R}^8$.

Density: $\pi^4 / 384 \sim 0.2537$.

Fields Medal 2018 for this proof.

Where Does Time Fall?

Time is not a singularity.

Time is the SEQUENCE of singularities.

TIME 1: COSMOLOGICAL TIME (t)

$t = 0$: Big Bang (the $0/0$)

$t > 0$: evolution (the removable values)

Inflation: $a(0) = 1 = 0^0$ (creation)

Matter: $a(0) = 0 = 0^x$ (annihilation)

Transition (reheating) resolves the $0/0$.

TIME 2: PRIME TIME (γ_n)

The zeta zeros γ_n are frequencies of prime oscillations.

$\psi(x) = x - \sum_n x^{\{1/2+i\gamma_n\}}/(1/2+i\gamma_n) + \dots$

$\gamma_1 = 14.1347$, $\gamma_2 = 21.0220$, ...

Prime time is ORDERED: each tick is a zero crossing.

TIME 3: MODULAR TIME (τ)

$\Delta(\tau) = q * \prod (1-q^n)^{24}$

The Fourier coefficients $\tau(n)$ are the 'moments'.

$\tau(1)=1$, $\tau(2)=-24$, $\tau(3)=252$, $\tau(4)=-1472$

The $0/0$: Δ has a zero at infinity, but $\tau(n)$ is nonzero.

Simplest Forms from Removable Singularities

Each removable singularity has an IRREDUCIBLE CONTENT:

$\sin(0)/0 = 1$ [L'Hopital 1696]
 $(e^x - 1)/x$ at $0 = 1$ [Newton 1687]
 $\log(1+x)/x$ at $0 = 1$ [Leibniz 1684]
 $(1 - \cos x)/x^2$ at $0 = 1/2$ [Taylor 1715]
 $(a^x - 1)/x$ at $0 = \log(a)$ [Euler 1748]
 x^x at $0 = 1$ [Combinatorics]
 $0! = 1$ [Combinatorics]
 $\Gamma(1) = 1$ [Euler 1729]
 $\zeta(0) = -1/2$ [Riemann 1859]
 $\zeta(-1) = -1/12$ [Ramanujan 1913]

THE ESSENCE OF $0/0$:

$0/0 = f'(0)/g'(0)$ (L'Hopital's rule)
 = the ratio of how fast numerator vanishes
 to how fast denominator vanishes
 This is the SIMPLEST learning machine:
 UNKNOWABLE (singularity) $\rightarrow$ KNOWABLE (removable value)

The Universal Learning Structure

[Von Neumann 1932; Connes 1994; Penrose 2004; Atiyah 1990]

PREMISE: Every field has singularities.

THE 5-STEP LEARNING PROCESS:

1. IDENTIFY the singularity (the $0/0$)
2. CLASSIFY it (removable, pole, essential)
3. COMPUTE the removable value
4. CONNECT it to other singularities
5. BUILD the network (the learning structure)

MATHEMATICS:

$\zeta(0)=-1/2$, $\zeta(-1)=-1/12$, index theorem, Langlands

PHYSICS:

Big Bang $a(0)=0^0=1$, Hawking $T=1/(8\pi M)$,
Bekenstein $S=4\pi M^2$, Holography $\text{bulk}(0)=\text{boundary}(\text{finite})$

BIOLOGY:

Extinction: $\text{count} \rightarrow 0$, fossil record=trace,
Speciation: $0 \rightarrow 1$, evolution=sequence of removable values

ECONOMICS:

Crash: $\text{price} \rightarrow 0$, true value=removable,
Bubble: $\text{price} \rightarrow \infty$, equilibrium=supply=demand

PHILOSOPHY:

Zeno: $0/0$ of motion, integral=whole,
Russell: self-reference, type hierarchy=solution

E8 as the Structure of Prime Time

E8 EXPONENTS: [1, 7, 11, 13, 17, 19, 23, 29]

These are 1 (identity) + first 7 primes after 5.

THE MISSING PRIMES: 2, 3, 5

These generate $A_3 = SU(4)$, $\dim=15$, $\text{rank}=3$.

$SU(4)$ exponents: 1, 2, 3.

THE HIERARCHY:

$SU(2)$: exponents 1. Primes: {2}

$SU(3)$: exponents 1,2. Primes: {2,3}

E8: exponents 1,7,11,13,17,19,23,29. Primes: {7-29}

E8 SKIPS the foundational primes 2,3,5.

The gap between $SU(n)$ and E8 is the COMPACTIFICATION.

E8 TIME DIRECTIONS:

8 axes, each with a PRIME FREQUENCY (the exponent).

Period = $2\pi / \text{frequency}$.

Product of periods = $(2\pi)^8 / (1 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29)$
= 0.011264

THE UNIVERSAL LEARNING MACHINE:

8 axes of knowledge, each with a prime frequency,
whose product gives the TOTAL KNOWLEDGE.

THE 0/0 OF E8:

Bosonic (112) + Fermionic (128) = 240 roots

The 0/0 is the UNIFICATION of bosons and fermions.

The removable value is the E8 VOLUME.

Computed with mpmath (30 digits). All numerical results verified.

E8 data: Conway-Sloane 1999, Viazovska 2017, Adams 1996.